Silvio Fanzon

Contact

Full Name	Silvio Fanzon
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Languages	English, Italian

Research Areas

Inverse Problems, Optimization, Numerical Analysis, Mathematical Imaging, Machine learning, PDEs, Optimal Transport, Calculus of Variations, Materials Science

Education

2014 – 2018 **PhD in Mathematics**, University of Sussex, UK
Thesis: *Geometric patterns and Microstructures in the study of Material Defects and Composites*
Grade: Pass with no corrections, Advisor: Prof. Mariapia Palombaro

2012 – 2014 **MSc in Mathematics**, Sapienza University, Italy
Thesis: *A variational approach to topological singularities in two-dimensions*
Grade: 110/110 Cum Laude, Advisor: Prof. Marcello Ponsiglione

2008 – 2011 **BSc in Mathematics**, Sapienza University, Italy
Thesis: *The isoperimetric problem*
Grade: 110/110 Cum Laude, Advisor: Prof. Annalisa Malusa

Professional Experience

04/23 – **Lecturer in Applied Mathematics**
Department of Mathematics
University of Hull, UK

06/22 – 03/23 **Faculty member (University Assistant)**
Department of Mathematics & Scientific Computing
University of Graz, Austria

04/18 – 10/21 **Postdoctoral Researcher**
Department of Mathematics & Scientific Computing
University of Graz, Austria

09/14 – 03/18 **Associate Tutor**
Department of Mathematics
University of Sussex, UK

Publications

Authors are in alphabetical order, denoting equal contribution

Peer-Reviewed Journal Articles

[1] *Asymptotic linear convergence of Fully–Corrective Generalized Conditional Gradient methods*
Mathematical Programming, 2023 (forthcoming)
(with K. Breides, M. Carioni, D. Walter)

[2] *A generalized conditional gradient method for dynamic inverse problems with optimal transport regularization*
Foundations of Computational Mathematics, Online first, 2022
(with K. Breides, M. Carioni, F. Romero)

[3] *A superposition principle for the inhomogeneous continuity equation with Hellinger-Kantorovich-regular coefficients*
Communications in Partial Differential Equations, 47(10):2023–2069, 2022
(with K. Breides, M. Carioni)

[4] *On the extremal points of the ball of the Benamou–Brenier energy*
Bulletin of the London Mathematical Society, 53(5):1436–1452, 2021
(with K. Breides, M. Carioni, F. Romero)

[5] *An optimal transport approach for solving dynamic inverse problems in spaces of measures*
ESAIM: Mathematical Modelling and Numerical Analysis, 54(6):2351–2382, 2020
(with K. Bredies)

[6] *Uniform distribution of dislocations in Peierls–Nabarro models for semi-coherent interfaces*
Calculus of Variations and Partial Differential Equations, 59(4):141, 2020
(with M. Ponsiglione, R. Scala)

[7] *Derivation of Linearised Polycrystals from a two-dimensional system of edge dislocations*
SIAM Journal on Mathematical Analysis, 51(5):3956–3981, 2019
(with M. Palombaro, M. Ponsiglione)

[8] *Optimal lower exponent for the higher gradient integrability of solutions to two-phase elliptic equations in two dimensions*
Calculus of Variations and Partial Differential Equations, 56(5):137, 2017
(with M. Palombaro)

[9] *A Variational Model for Dislocations at Semi-coherent Interfaces*
Journal of Nonlinear Science, 27(5):1435–1461, 2017
(with M. Palombaro, M. Ponsiglione)

Conference Proceedings & Others

[10] *Novel Convex Hybrid Image and Motion-Field Reconstruction*
"2021 ISMRM & SMRT Annual Meeting & Exhibition", 15-20 May 2021, Vancouver, Canada
(with K. Breides, A. Martin, I. Middelhoff, M. Schlögl, R. Stollberger)

[11] *Geometric patterns and microstructures in the study of material defects and composites*
Doctoral Thesis (PhD), University of Sussex, 2018

Research impact: Projects and Funding

2020 – 2021	Participation in FWF Research Project lead by Kristian Bredies *P 29192: Regularization Graphs for Variational Imaging*
2018 – 2020	Participation in FWF Research Project lead by Kristian Bredies *PIR-27: Mathematical methods for motion-aware medical imaging*
2014 – 2018	Full-time PhD Studentship from the University of Sussex and fee waiver
2014 – 2018	Research Grant of £5800 from the University of Sussex
2014 – 2018	Travel support to attends events from Carnegie Mellon University (US), Oxford University (UK) University of Warwick (UK), SISSA University (Italy), National Research Council (Italy)

Teaching experience

Lecturer

2023/24	**Statistical Models**, Y2 BSc Math	University of Hull, UK
	Numbers, Sequences & Series, Y1 BSc Math	
	Differential Geometry & Topology, Y3 BSc Math	
2022/23	**Inverse Problems** (Practical), Y2 MSc Math	University of Graz, Austria
	Analysis 3 (Practical), Y2 BSc Math	
2020/21	**Calculus of Variations**, Y3 BSc Math	
2019/20	**Advanced Functional Analysis** (Practical), Y1 MSc Math	

Teaching Assistant

2017/18	**Geometry 1**, Y1 BSc Math	University of Sussex, UK
2016/17	**Analysis 1**, Y1 BSc Math	
	Mathematics Demystified, Y1 BSc Math	
	Introduction to Probability, Y1 BSc Math	
2015/16	**Probability and Statistics**, Y2 BSc Math	
	Introduction to Probability, Y1 BSc Math	
2014/15	**Probability and Statistics**, Y2 BSc Math	
2012/13	**Ordinary Differential Equations**, Y2 BSc Math	Sapienza University, Italy

Mentoring

2023/24 **David Awuku**, MSc Mathematics — University of Hull, UK
Thesis: *Entropic Regularization of Optimal Transport*

Presentations

denotes invited presentation

Oral Presentations

2023 **Seminar, Department of Mathematics**
University of Sussex, UK, 13 Mar 2023

2022 **Sussex Mathematics Seminar**
University of Sussex, UK, 3 Nov 2022

Seminar, Department of Mathematics
Heriot-Watt University, UK, 13 Sep 2022

Seminar, Department of Mathematics & Scientific Computing
University of Graz, Austria, 18 Feb 2022

2021 **SIMAI 2020-2021**
University of Parma, Italy, 30 Aug - 3 Sep 2021

2019 **M.A.G.A. Days (Monge-Ampère et Géométrie Algorithmique)**
Laboratoire de mathematiques d'Orsay, France, 20-21 Nov 2019

1st Austrian Calculus of Variations Day
University of Vienna, Austria, 17-18 Oct 2019

ICCOPT: 6th International Conference on Continuous Optimization
Technical University Berlin, Germany, 3-8 Aug 2019

2018 **Topics in Nonlinear Analysis: Calculus of Variations and PDEs**
University of Lisbon, Portugal, 10-12 Oct 2018

Seminar, Department of Mathematics & Scientific Computing
University of Graz, Austria, 31 Jan 2018

2017 **XXVII National meeting of Calculus of Variations**
CIRM, Levico Terme, Italy, 6-10 Feb 2017

2016 **Working Seminar on Calculus of Variations**
Sapienza University, Italy, 19 Dec 2016

Poster Presentations

2021 **ITN TraDe-OPT Winter School**
Online event, 15-19 Feb 2021

2016 **Hysteresis, Avalanches and Interfaces in Solid Phase Transformations**
University of Oxford, UK, 19-21 Sep 2016

PIRE-CNA. New Frontiers in Nonlinear Analysis for Materials
Carnegie Mellon University, United States, 2-10 Jun 2016

Research stays

2022	University of Sussex, UK, 1-22 Jul and 27 Oct - 5 Nov Invited by Filippo Cagnetti
	Heriot-Watt University, UK, 12-16 Sep Invited by Panagiota Birmpa
2019	Sapienza University, Italy, 15-19 Apr and 8-17 Jul Invited by Marcello Ponsiglione
2018	Sapienza University, Italy, 17-21 Dec Invited by Marcello Ponsiglione
	University of Graz, Austria, 31 Jan - 2 Feb Invited by Kristian Bredies

Reviewer activity

SIAM Journal on Mathematical Analysis
SIAM Journal on Imaging Sciences
Mathematics in Engineering
Numerische Mathematik
Inverse Problems and Imaging

IT Skills

Code	Python, Matlab, R, C
Teach	LaTex, Canvas, Panopto, Mathematica, MS Office, MS Teams
Web	HTML, CSS, JavaScript